

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CE)

May 2012

CE 208: OBJECT ORIENTED CONCEPTS USING VISUAL C++ PROGRAMMING

Date: 16.05.2012, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1 Answer the following Questions.

- | | |
|---|--------------|
| (a) Constructor can not return a value. State True or false. Justify | 01 |
| (b) Explain features of Object Oriented Programming. | 03 |
| (c) Define: Run time binding, Pure Virtual Function | 02 |
| (d) Which of the following ways are legal to access a class data member using this pointer? | 01 |
| a) this.x | b) (*this).x |
| c) *this.x | d) *(this.x) |

Q - 2 Answer the following Questions.

- | | |
|---|----|
| (a) Explain Call by Reference and Return by Reference with example. | 04 |
| (b) What is Type Conversion? Why it is needed in object oriented programming? Explain any two type conversion technique with example. | 05 |
| (c) Explain concept of pointers to objects with example. | 02 |
| (d) Explain concept of Friend Function with example. | 02 |
| (e) If the visibility is private then which of the following is true. | 01 |
| a) Private remains private in derived | |
| b) Protected remains protected in derived | |
| c) Public remains public in derived | |
| d) None of these | |

OR

Q - 2 Answer the following Questions.

- | | |
|--|----|
| (a) Explain Copy Constructor and inline function with example | 04 |
| (b) What is Inheritance? List and Explain different types of inheritance with diagram. | 05 |
| (c) Explain Concept of Reference variable with example. | 02 |

- (d) What is Operator Overloading? List out the operators which are not overloaded. 02
- (e) What will be the output of the following program? 01

```
#include<iostream.h>
```

```
int main() {
```

```
    float x=5,y=2;
```

```
    int result;
```

```
    result=x % y;
```

```
    cout<<result;
```

```
    return 0; }
```

Q - 3 Answer the following Questions.

- (a) What is MFC? Explain WinMain Function with arguments. 03
- (b) State True or false. 01
We can use loop and nesting of if in inline function.
- (c) Write a program to enter the information of n number of various cars and display it using multilevel inheritance. 05
- (d) Write a program of operator overloading that overload arithmetic assignment operator. 05

OR

Q - 3 Answer the following Questions.

- (a) Explain Create function and PaintDC class in MFC. 03
- (b) State True or false. 01
Friend function can be friend of many classes at the same time.
- (c) Write a program that illustrates the hierarchical inheritance in creating two new class called savings account and current account from the base class account. 05
- (d) Write a program to enter information of an employee and display it by using the run time binding technique. 05

SECTION – II

Q - 4 Answer the following Questions.

- (a) Define UML Diagrams. List out the different types of UML Diagrams. 02
- (b) Define Collaboration and Sequence Diagrams. Briefly explain about UML interaction diagrams with suitable example. 03
- (c) Differentiate Object Oriented Analysis and Object Oriented Design. 02

Q - 5 Answer the following Questions.

- (a) Write steps and a program MFC program to create a window of desired size. 04

- (b) Compare Aggregation and Association with example. 04
- (c) Differentiate Followings: 06
- 1) Conditional Behavior and Parallel Behavior
 - 2) Abstract Class and Interface

OR

Q - 5 Answer the following Questions.

- (a) What is a state diagram? Explain the characteristics of the system that can be identified by examination of the state diagram of that System. 04
- (b) Write steps and a MFC program to insert menu in particular window. 04
- (c) Why DFD is not included in UML? What is the difference between DFD, Activity Diagram and Flow Chart? 06

Q - 6 Answer the following Questions.

- (a) Draw the use-case diagram for Banking System 05
- (b) Categorize the following relationship into generalization, aggregation and association. 05
- Files contain records
 - A drawing object is text, a geometrical object or group.
 - Modems and keyboards are input/output devices
 - Object classes may have several attributes.
- (c) What will be the output of the following programs? 04

<pre>#include <iostream.h> class easy { private: int i; public: easy(int num) { i = num; } void operator++(int) { cout<<"First";} void operator++(void) { cout<<"Hi"; } }; int main() { easy obj(5); obj++; return 0; }</pre>	<pre># include <iostream.h> void junk(int, int*); int main() { int i=6, j = -4; junk(i, &j); cout<<"i="<<i<<"j="<<j; return 0; } void junk(int a, int *b) { a = a*a; *b = *b * *b; }</pre>
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OR

Q - 6 Answer the following Questions.

- (a) Draw the use-case diagram for conduct of Entrance Exam and Admission Process for Engineering Students in Gujarat. **05**
- (b) Draw Activity diagram for Online Shopping System. **05**
- (c) What will be the output of the following programs? **04**

```
#include <iostream.h>

int main() {
    void fun1(void);
    void fun2(void);
    fun1();
    fun2();
    return 0; }

void fun1() { cout<<"Hello"; }
void fun2() { cout<<"to you"; }
```

```
#include <iostream.h>

class value {
    int i; float f;
    value() { i = 12; f = 3.14; }
    cout<<i<<"", "<<f; }
};

int main() { value v1; }
```
